GISTDA GEOHACKATHON 2026

Pixels 2 Metro Policy

A Geospatial Analysis Framework and Methodological Toolkit for Revealing Hidden Challenges in Bangkok's Informal Settlements

Community records + building evidence + Satellite image



Khlong Toei, Bangkok / Photograph: Unequal Scenes

Main message

Data used on local level of metropolitan planning and management often is outdated, fragmented and biased – real challenge for modern local urban governance. Metropolises change every day.

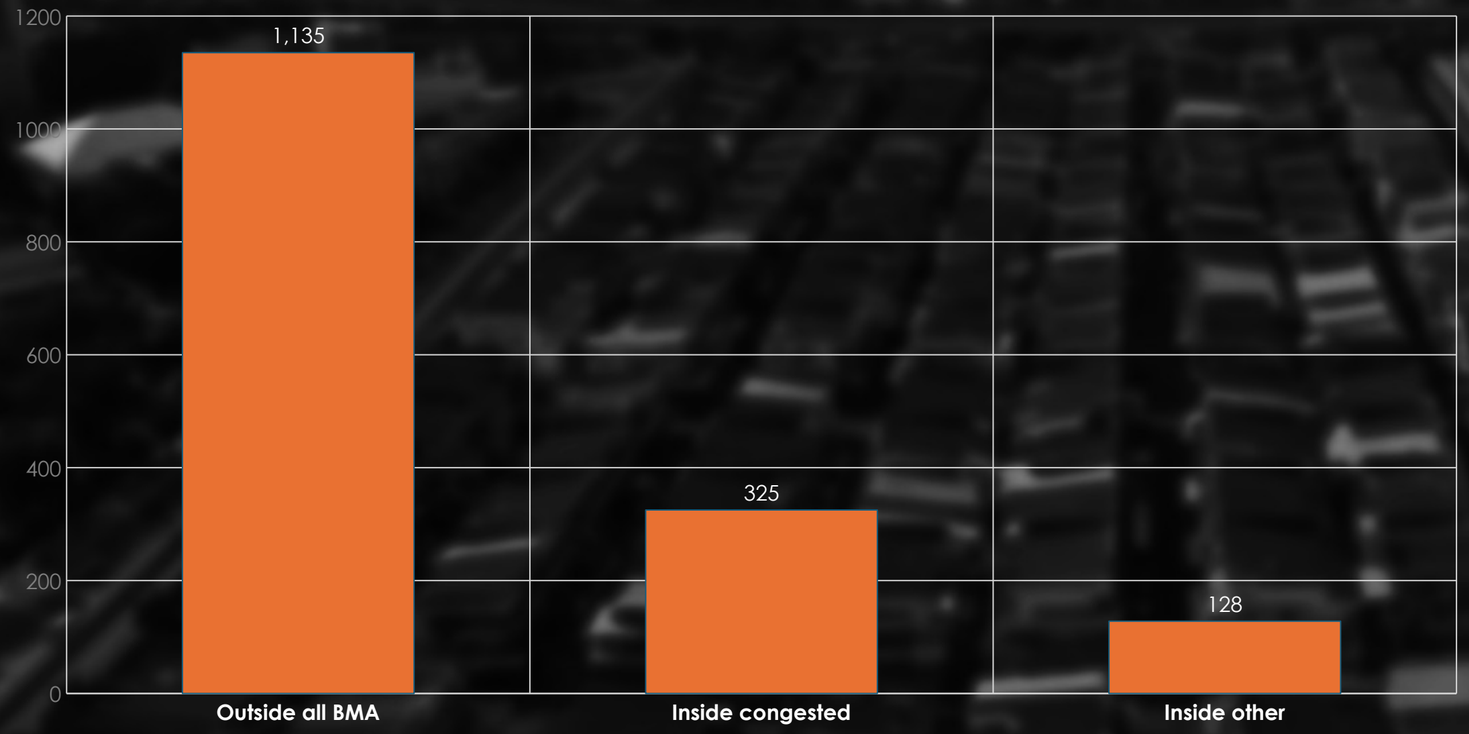
Pixels2MetroPolicy

Interactive tool for metropolitan planning and management
- identifies slum and sensitive neighborhoods with slum-like risks.
- aggregates everyday data from statistic sources and social media, scores risks and makes ranking of problems in identified neighboorhoods.
- proposes range of policy toolkits (from the World's case studies) to be applied by decision of metropolitan governance body

Identify
analyze
Understand
Communicate
Act

The city’s housing records do not line up

NHA low-income locations and BMA community boundaries describe different sets of places.



1,588 NHA points compared

A cross-register mismatch. Different dates, definitions and boundaries can explain the gap.

71%

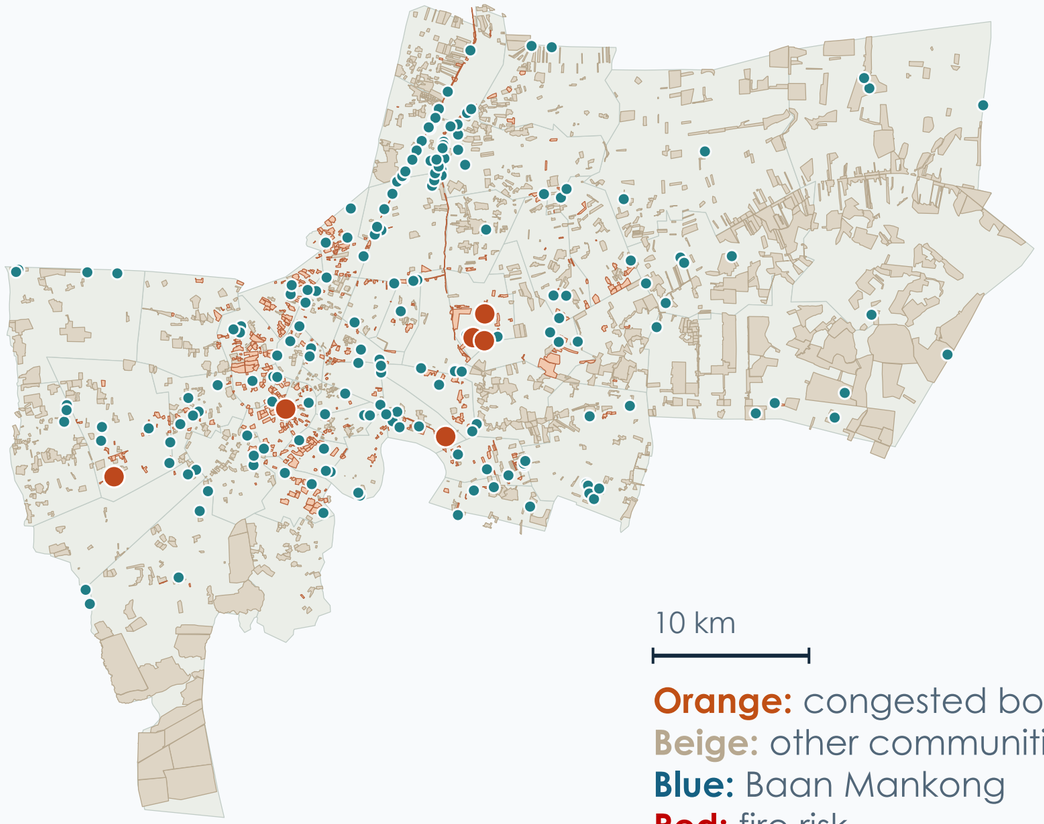
of NHA points fall outside BMA community polygons

Planning use

Reconcile source dates and boundaries
before treating a record gap as a new settlement.

Community boundaries, housing projects and risk

The layers connect settlement geography with housing programmes and assessed fire risk.



626

congested-community polygons

177

Baan Mankong locations

6 high / critical

high-risk communities

Project points show locations, not project boundaries.

Citywide screening produces a review queue

Local satellite segmentation and the citywide building model use separate inputs and serve different scales.



THEOS-2 roof extraction

2,284 polygons across two scenes

SEPARATE CITYWIDE PIPELINE

1.80M footprints

Open building records clipped to the study area

123,917 grid cells

Building patterns summarized at 100 m resolution

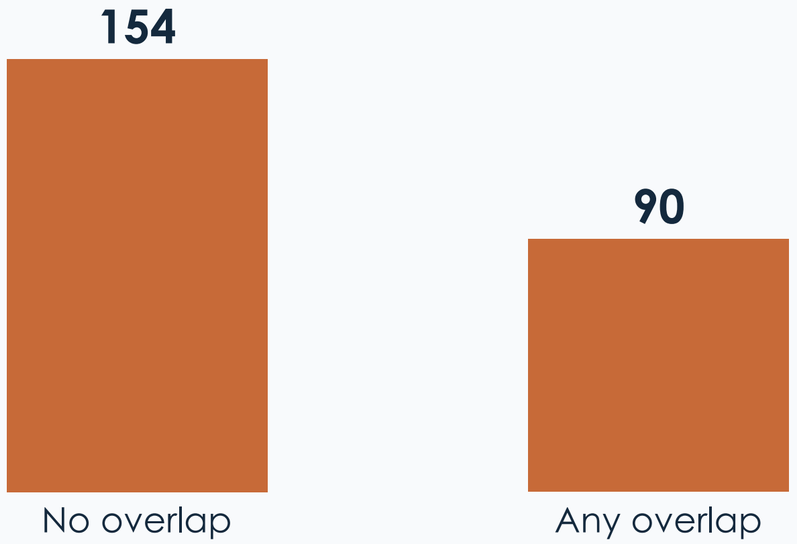
244 review areas

Clusters retained after register and distance filters

Output supports a survey queue; validation is pending.

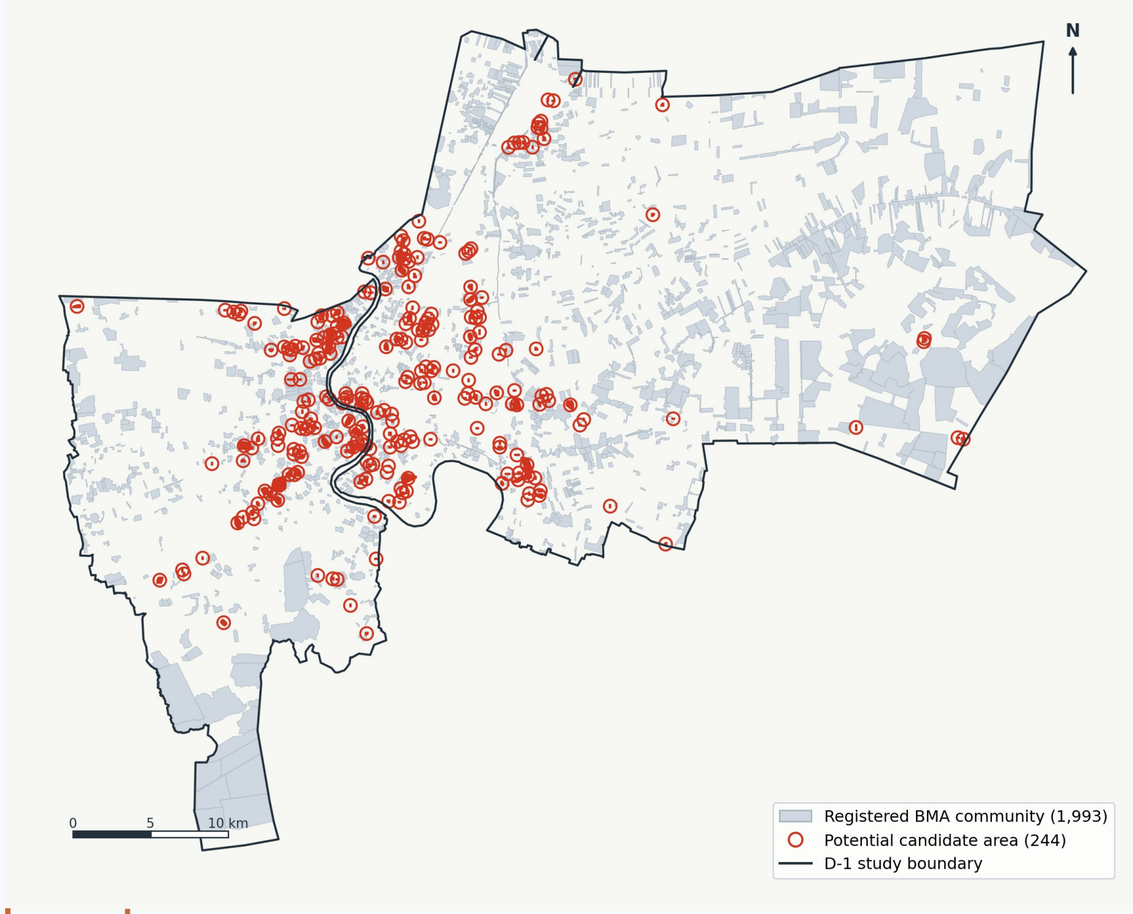
244 areas to check, with boundaries to reconcile

The final polygon check separates areas with no register overlap from areas needing boundary review.



154 areas: inspect evidence beyond the register
90 areas: reconcile overlapping boundaries

Use together with existing registered communities.



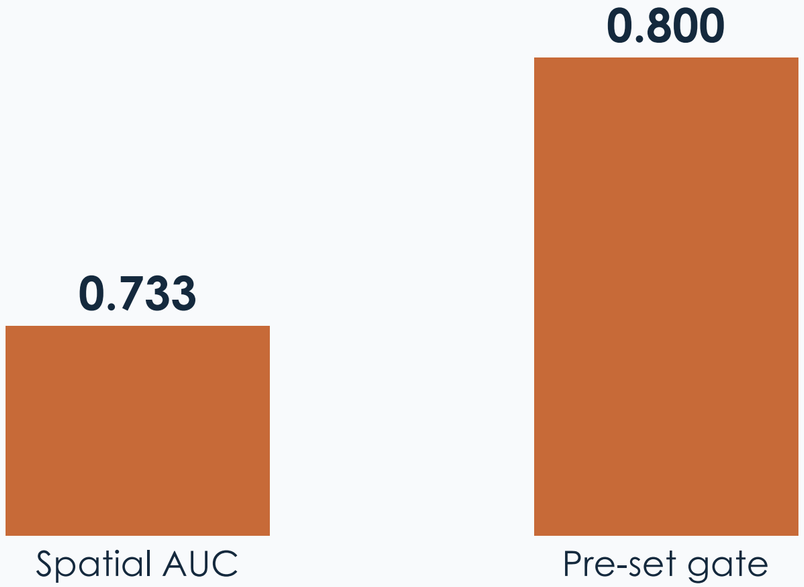
Unranked screening output. Zero independently confirmed settlements.

The audit changed what we could claim

Scores show resemblance to patterns in the training labels. Independent evidence must establish local conditions.
Spatial validation limits classification; a corrected join changes how buildings count inside communities.

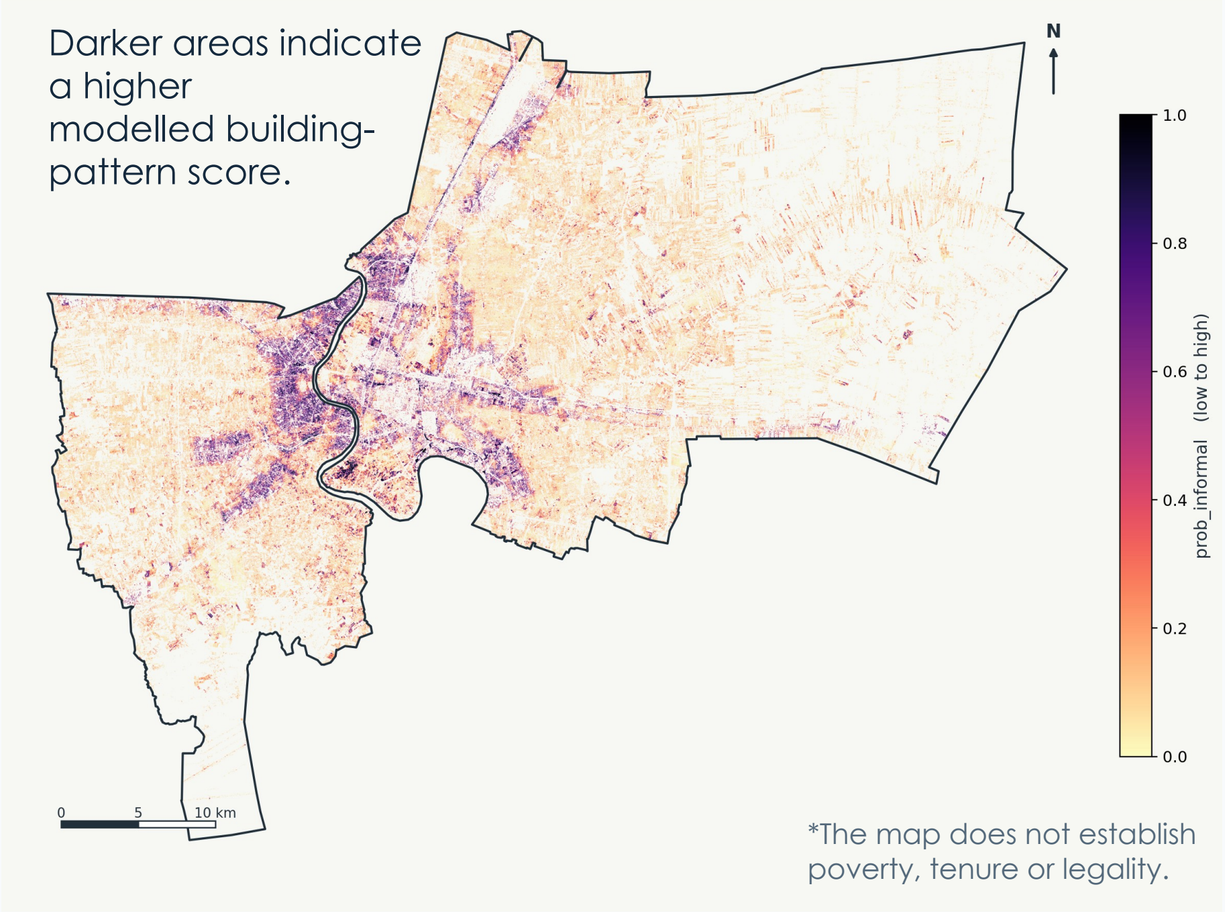
MODEL PERFORMANCE

Spatial AUC 0.733



Below the gate for classification

AUC measures discrimination across held-out districts.
Registration-derived labels do not establish settlement status.

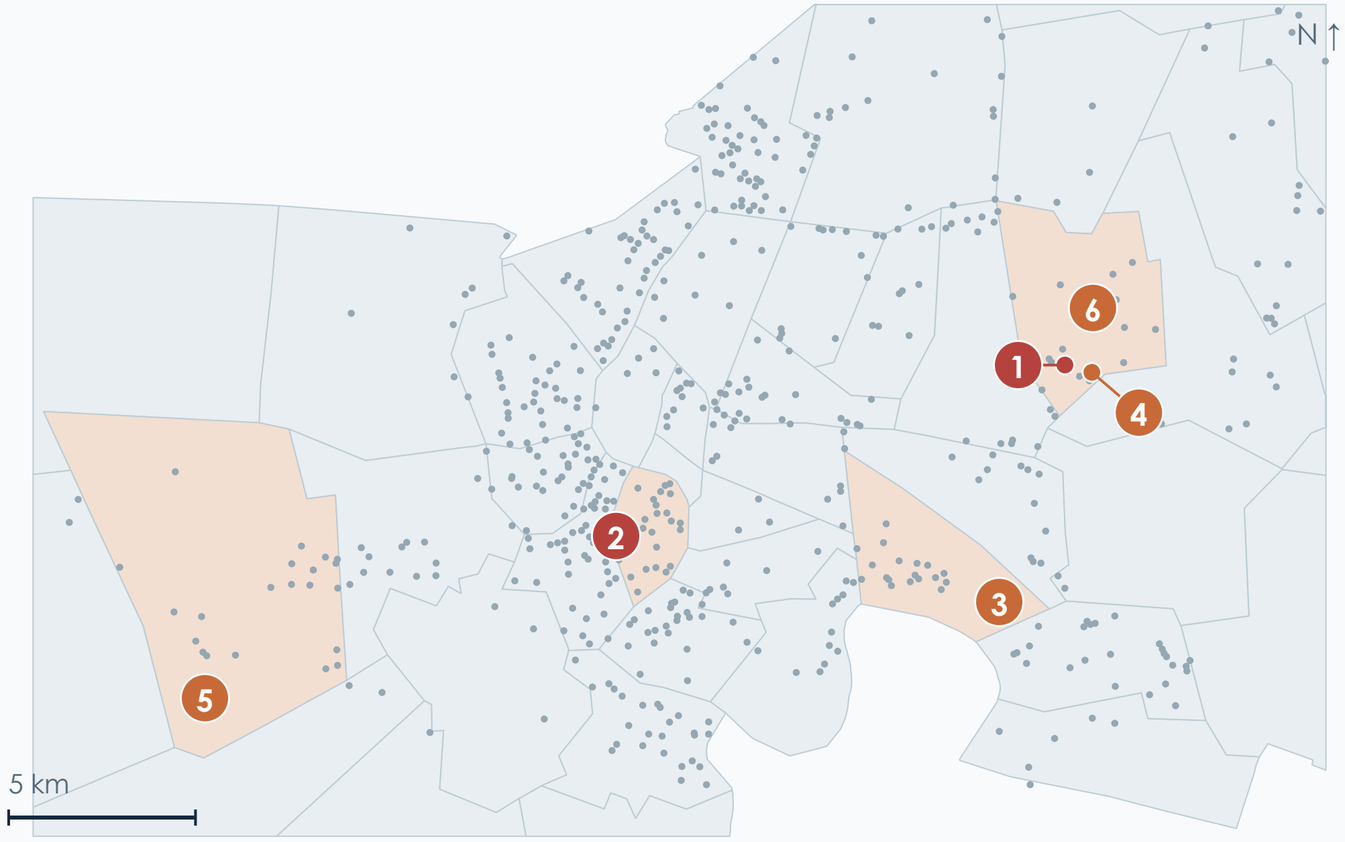


*The map does not establish poverty, tenure or legality.

A counting correction, not newly detected buildings. Independent settlement validation remains open.

SEA evidence: fire risk by community

Assessment year 2568: 2 critical and 4 high-risk communities. Three are in Wang Thong Lang.



1. Nom Klao

Wang Thong Lang / Critical

2. San Chao A Niao

Khlong San / Critical

3. Rim Khlong Phra Khanong

Khlong Toei / High

4. Rim Khlong Lat Phrao

Wang Thong Lang / High

5. Mi Suwan Phatthana

Bang Khae / High

6. Surao Don Sakae

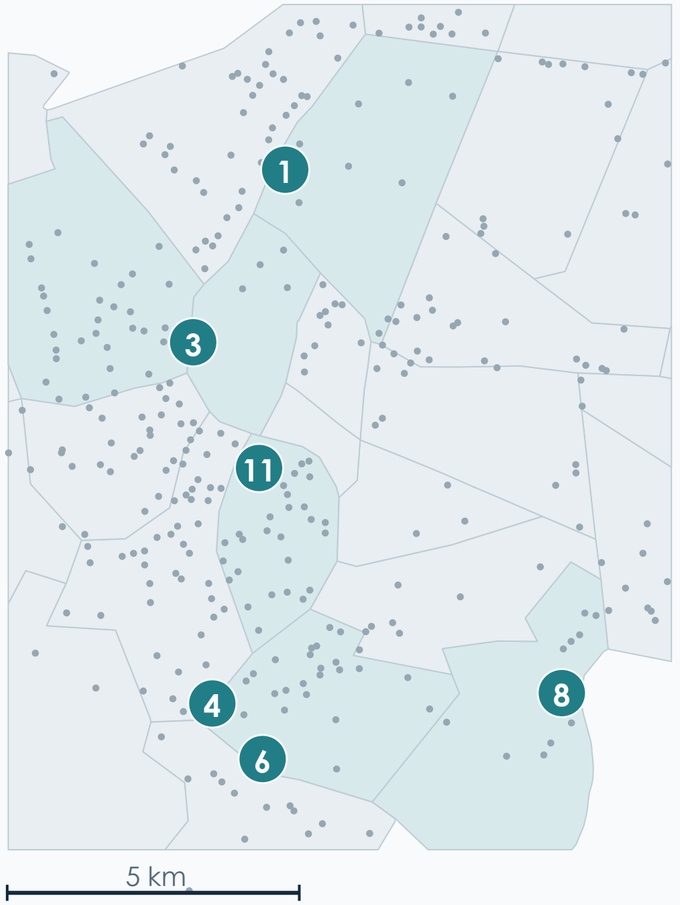
Wang Thong Lang / High

Grey points: community locations. Coloured districts contain the six assessed communities.

SEA evidence: communities outside flood walls

More than 11 registered informal communities, 320 households and about 1,070 people outside the flood walls.





01	Ratchapha Thapthim Ruam Chai
02	Thewarat Kunchon
03	Tha Wang
04	Mata Nuson
05	Behind Charoen Krung Hospital
06	Soi Wat In Ban Chong
07	Wat Bang Khlo Nok
08	Rice Mill, Rama III Road
09	Wat Dusit Nimit Mai
10	End of Soi Charoen Krung 29/2
11	Chang Nak–Saphan Yao

Six coordinate matches in the available file. Five locations remain unmatched or unconfirmed.

SEA evidence: household burdens

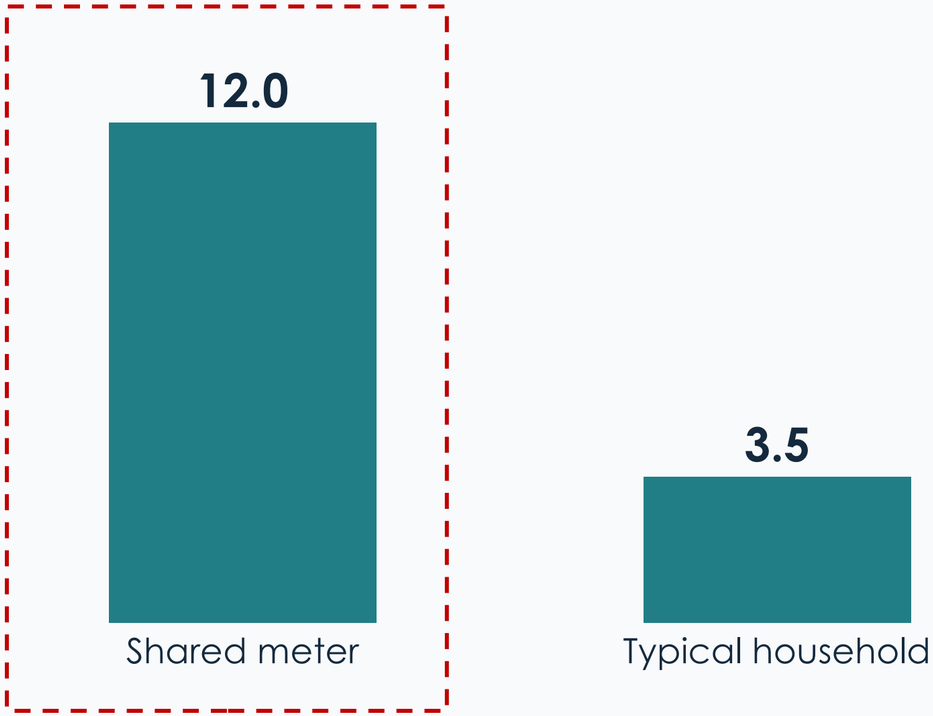
Limited income and shared utility meters can compound household vulnerability.

MONTHLY INCOME PER PERSON



Baht per person per month

ELECTRICITY COST PER UNIT



Bang Na case: about 3.4 times

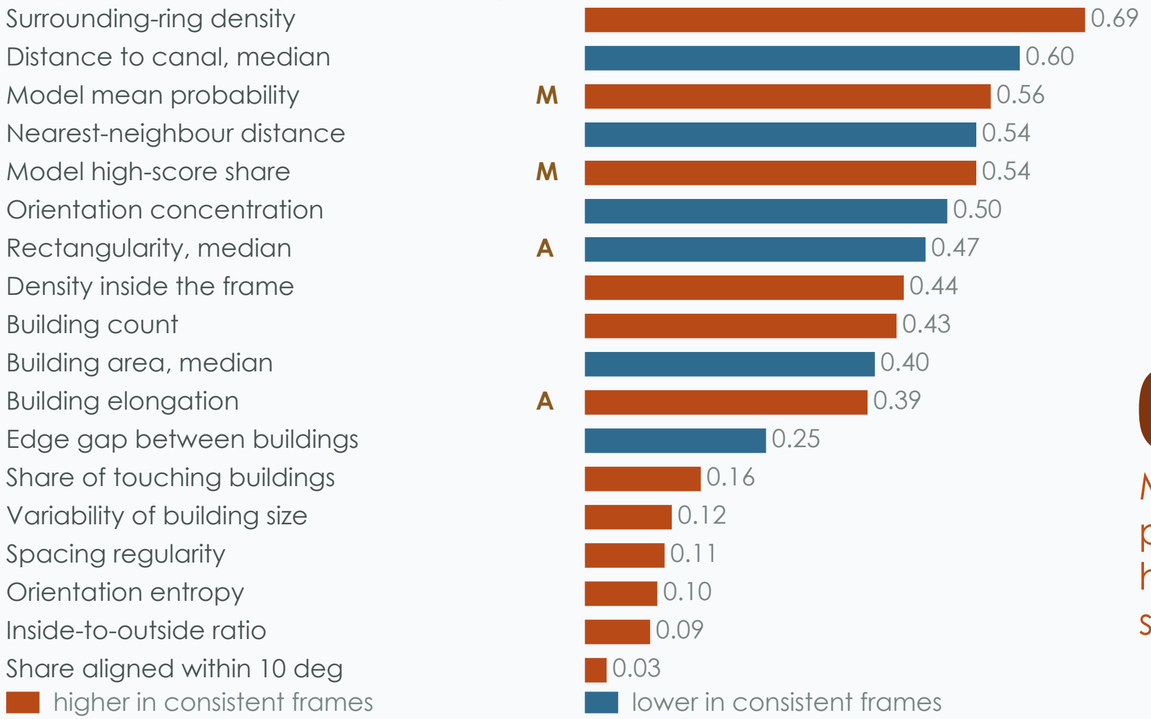
Historical source figures and one shared-meter case. Different populations and reference periods.

Density dominates the morphology diagnostics

Eighteen descriptors, measured across 40 frames holding 28,950 buildings, with 10 frames judged consistent against 20 inconsistent by one reviewer. Direction only, no interval.

0.98
Building count and inside density

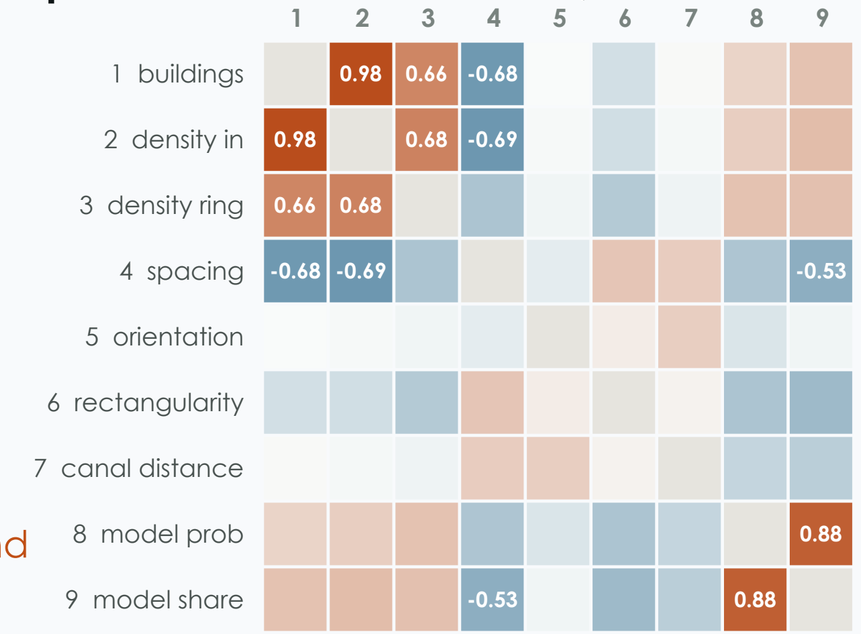
Separation, |AUC - 0.5| x 2 0 means no separation



M = the model's own output. A = known artefact of the footprint source.

0.88
Model probability and high-score share

Spearman rank correlation, n = 40 frames



Nine of the eighteen descriptors exist at frame level; the other nine are published only as summary rows. No missing values. Correlation is not predictive importance, not separation, and not causation.

Building count and density inside the frame correlate at 0.98, and both track the surrounding ring. Four of the five leading separators are density at different radii, and the fifth is a model input. None of them establishes settlement status.

Sources . morphology_diagnostic_table.csv, 18 rows, separation verified for every row; group_membership.csv, 40 frames x 9 descriptors, no missing values; phase5_results.json strata.

The evidence continues in our web app

Explore Bangkok community data

01 Choose a district

Locate communities in the area you want to examine.

02 Compare map layers

View community boundaries, SEA analysis, Segmentation analysis, Overlay mapping analysis

03 Read the local evidence

Connect the map with community information and the social and infrastructure analysis.

OPEN THE WEB APP

GEOHACKATHON 2026

Thank you

Pixels2MetroPolicy
Roman Pomazan
Pruethicheth Lert-Udompruksa
Olga Panina

Khlong Toei, Bangkok / Photograph: Unequal Scenes